



**CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION**

PHYSICS

UNIT 2 - PAPER 02

2 hours and 15 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of **NINE** questions.
2. Section A consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in this answer booklet.
3. Section B consists of **SIX** questions. Candidates must attempt **THREE** questions in this section, **ONE** question from **EACH** Module. Answers for this section must be written in the answer booklet provided.
4. All working **MUST** be **CLEARLY** shown.
5. The use of non-programmable calculators is permitted.

LIST OF PHYSICAL CONSTANTS

Speed of light in free space	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_0	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_0	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$
Elementary charge	e	=	$1.60 \times 10^{-19} \text{ C}$
The Planck's constant	h	=	$6.63 \times 10^{-34} \text{ J s}$
Unified atomic mass constant	u	=	$1.66 \times 10^{-27} \text{ kg}$
Rest mass of electron	m_e	=	$9.11 \times 10^{-31} \text{ kg}$
Rest mass of proton	m_p	=	$1.67 \times 10^{-27} \text{ kg}$
Acceleration of free fall	g	=	9.81 m s^{-2}
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$

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SECTION A

Attempt ALL questions. You MUST write in this answer booklet. You must NOT spend more than 30 minutes on this section.

1. (a) A student wishes to determine the electromotive force (e.m.f.), E , and internal resistance, r , of a battery. V is a voltmeter and R is a variable resistor. She sets up the circuit shown in Figure 1a and finds that it does not work.

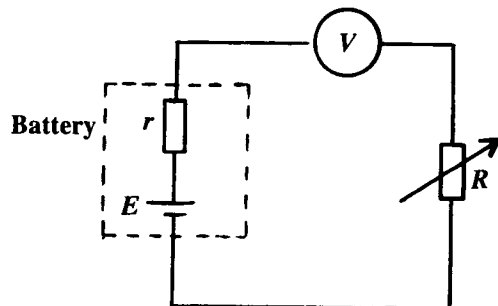


Figure 1a

- (i) Identify the problem affecting the circuit.

[1 mark]

- (ii) Redraw the circuit so that it will work.

[1 mark]

- (iii) State which readings should be taken.

[1 mark]

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- (b) The experimental quantities are related by the equation $\frac{I}{V} = \frac{I}{E} + \frac{r}{E} \frac{I}{R}$.

Figure 1b shows a graph of $\frac{I}{V}$ against $\frac{I}{R}$ for data obtained from the redrawn circuit.

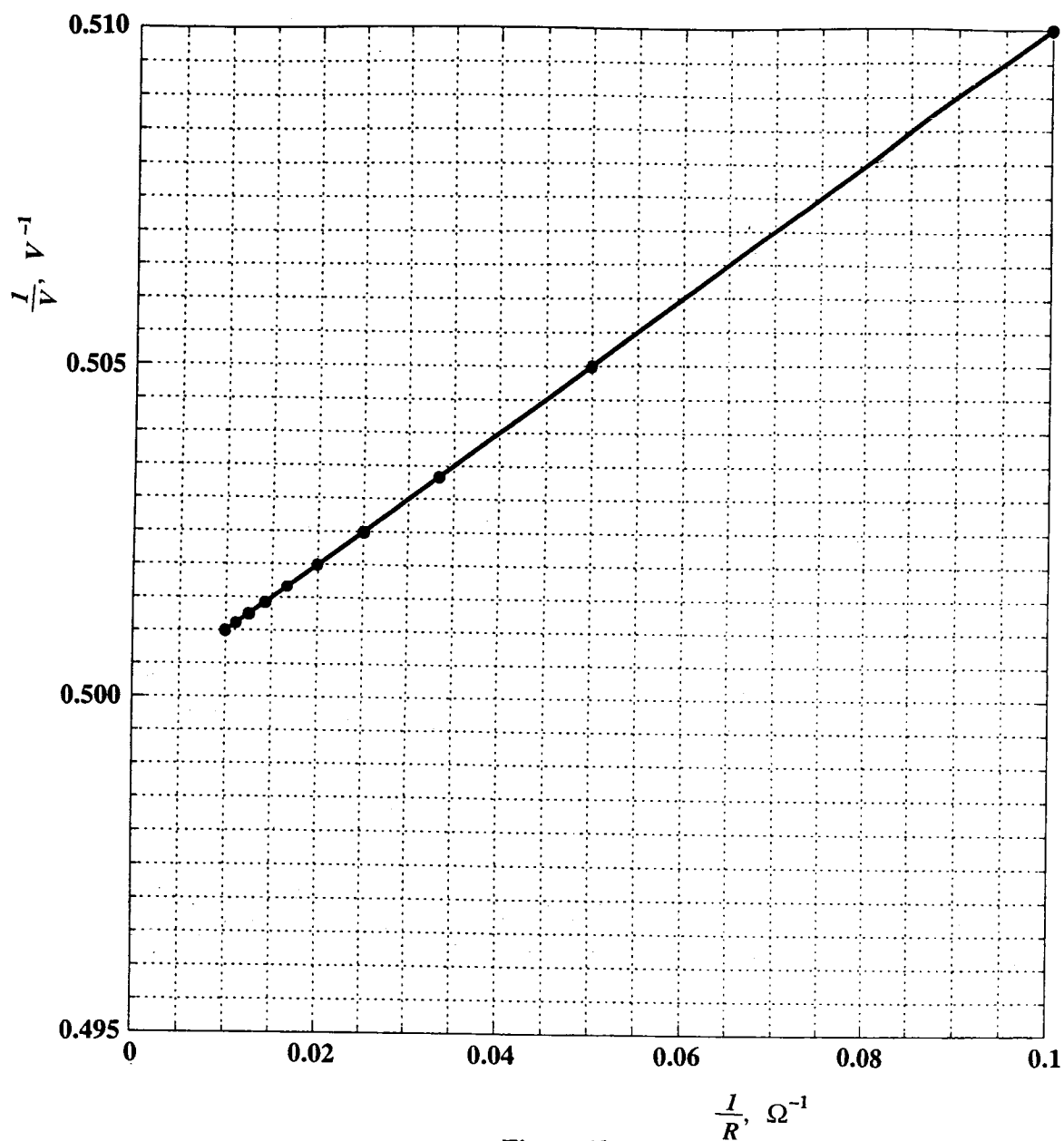


Figure 1b

Determine

- (i) a) the e.m.f., E , of the battery

[3 marks]

- b) the internal resistance, r , of the battery.

[3 marks]

- (ii) If the resistance of the voltmeter were comparable to that of R , how would this have affected the result?

[1 mark]

Total 10 marks

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2. (a) The markings on an old transformer are not shown. A student connects the circuit shown in Figure 2a in an attempt to determine the properties of the transformer.

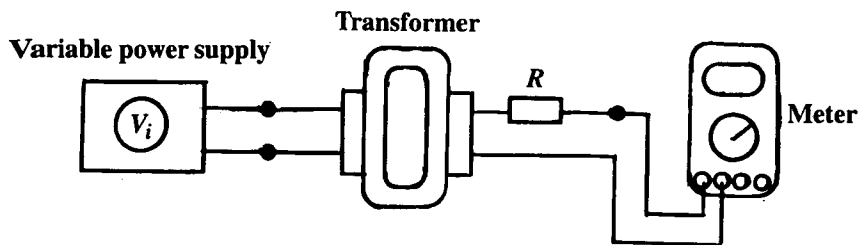
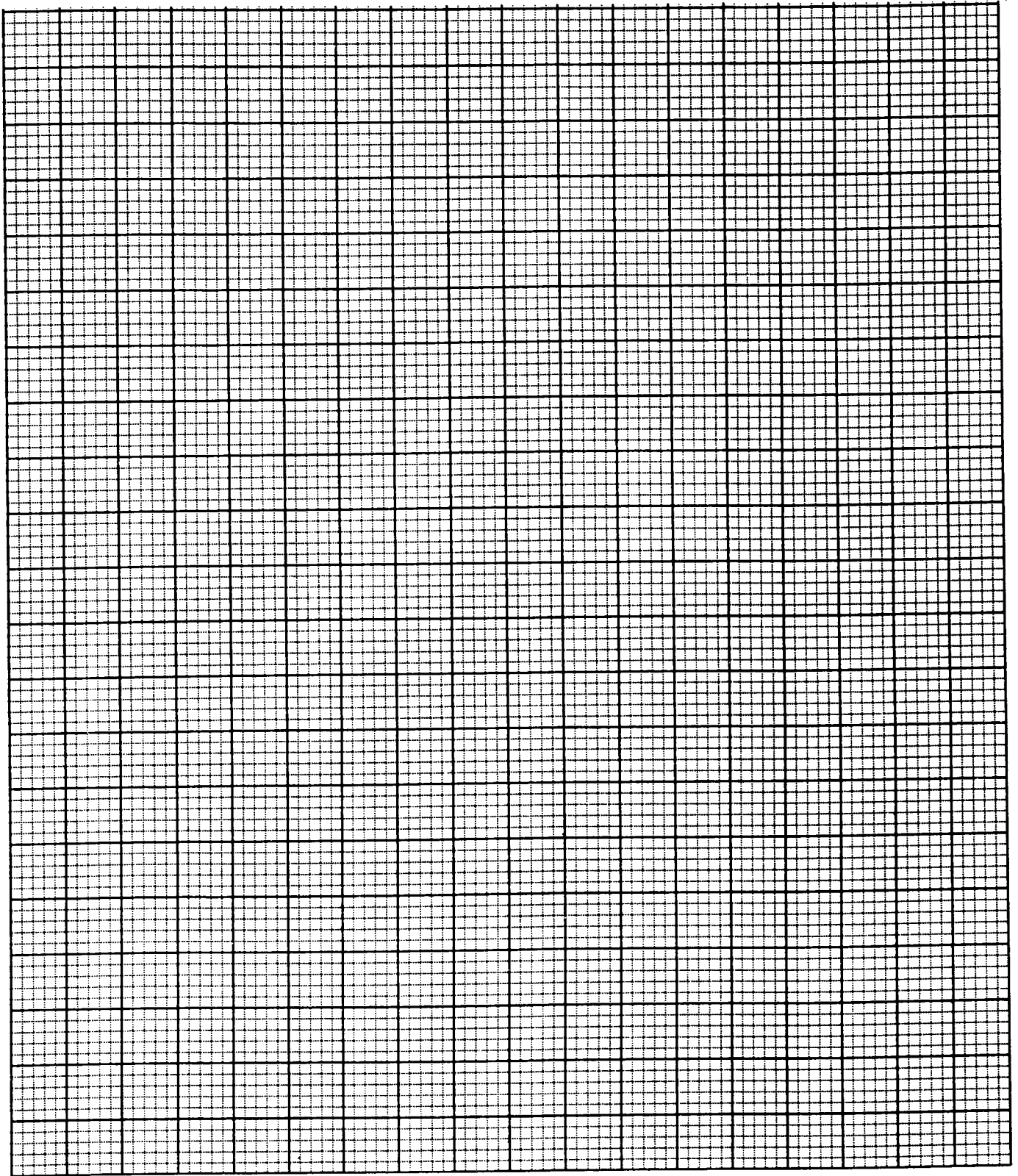


Figure 2a

The following data were obtained.

Input voltage, V_i (V)	240	200	160	140	120
Output voltage, V_o (V)	47.8	39.7	31.8	28.1	24.2

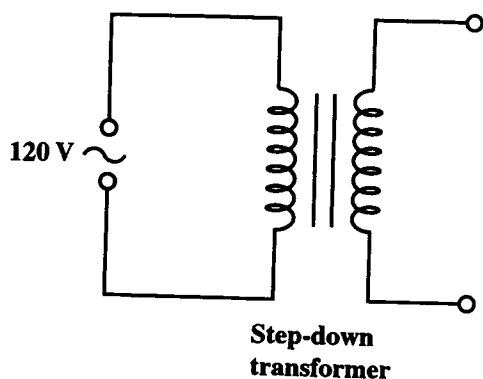
- (i) Plot a graph of input voltage, V_i , against output voltage, V_o , on the grid on page 7. [4 marks]
- (ii) Use the graph to determine the turns ratio of the transformer.
- _____
- _____
- _____
- _____
- [2 marks]
- (iii) If the number of turns in the secondary windings is quoted at 30 000, what is the number of turns in the primary windings?
- _____
- _____
- _____
- _____
- _____
- [1 mark]



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- (b) You need to produce an approximately 4 V DC output from an AC power supply whose output is shown in Figure 2b. You have available a transformer, a capacitor, a diode, a cathode ray oscilloscope and a resistor to represent the load.

- (i) On the diagram below, complete the design of a circuit to achieve this objective, using all of these components.



[1 mark]

- (ii) Figure 2b shows the output voltage from the AC power supply.

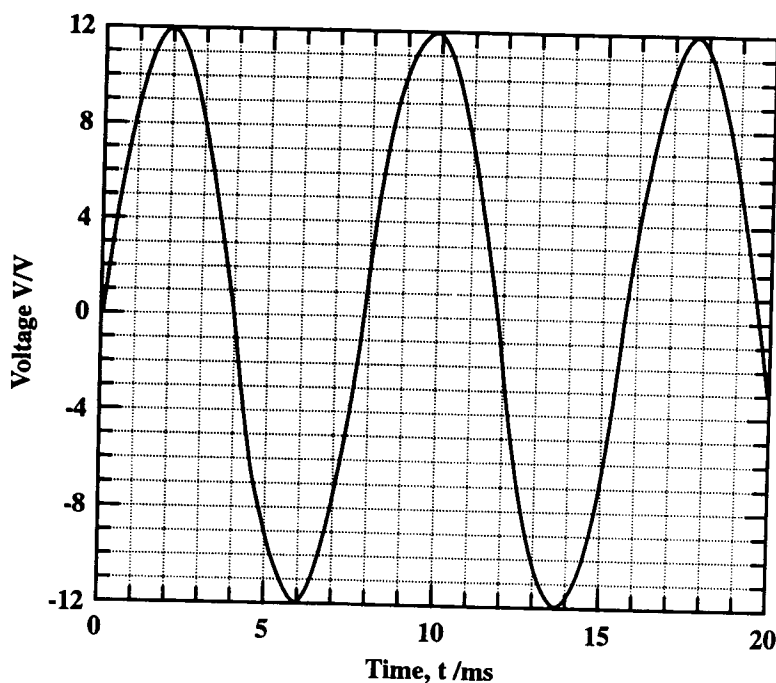


Figure 2b

On Figure 2b, sketch the output voltage that would be seen on the oscilloscope.
[2 marks]

Total 10 marks

- (b) Figure 3*b* shows a graph of V against $\frac{1}{\lambda}$ for the data obtained.

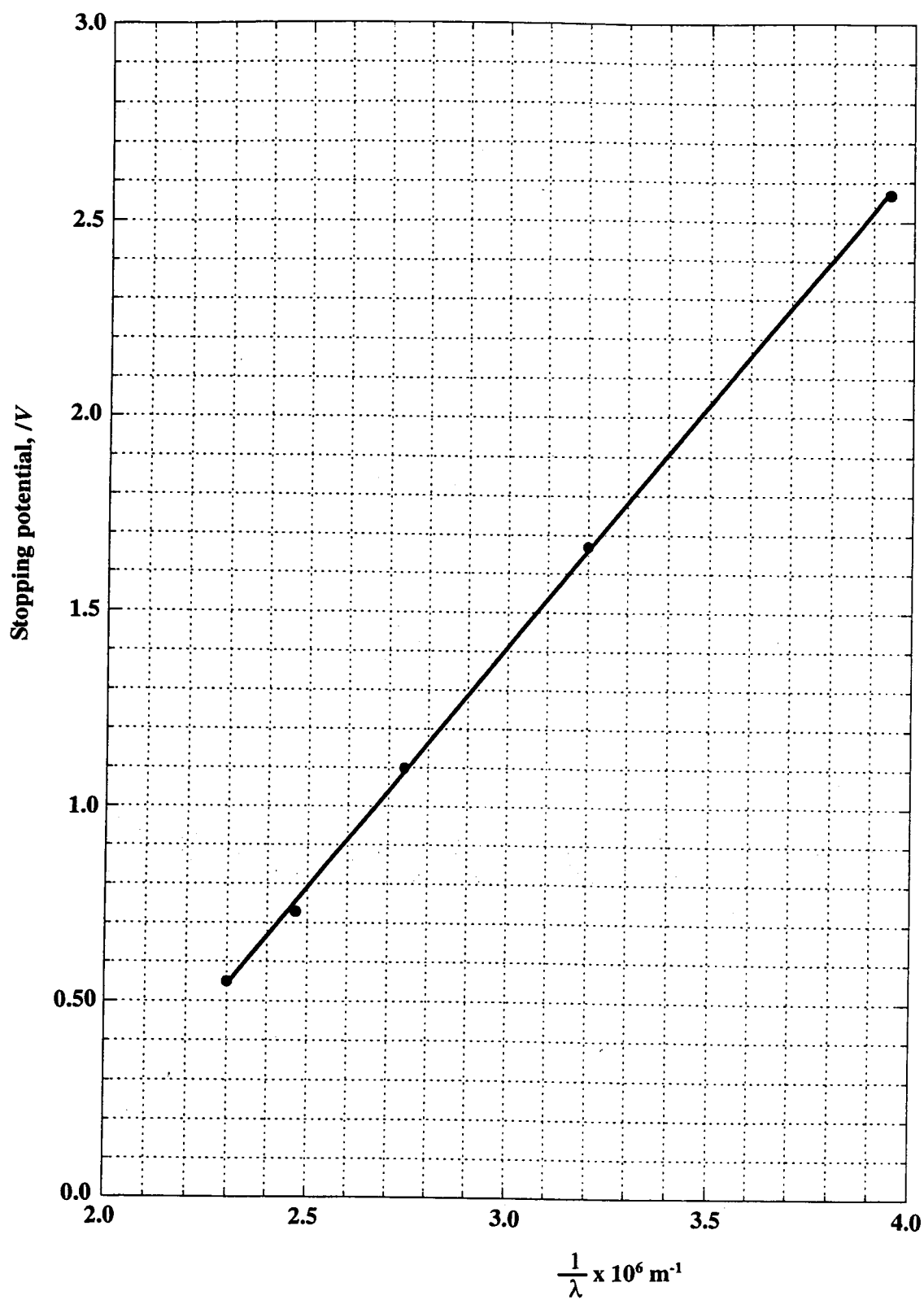


Figure 3*b*

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Use the graph to determine

- (i) Planck's constant

[3 marks]

- (ii) the cut-off wavelength λ_0 .

[3 marks]

Total 10 marks

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SECTION B

You must attempt **THREE** questions from this section. Choose **ONE** question **EACH** from Module 1, 2 and 3. You **MUST** write your answers in the answer booklet provided.

MODULE 1

Answer **EITHER** Question 4 **OR** Question 5.

4. (a) Two point charges have magnitudes of q_1 and q_2 and are separated by a distance r .
- (i) State **THREE** facts about the electrostatic force that exists between these point charges.
 - (ii) Write an expression for the electric field experienced by charge q_2 due to charge q_1 .
 - (iii) Figure 4a shows two negative point charges with equal charges.



Figure 4a

Copy the figure into your answer booklet and draw the resulting electric field lines. Indicate with an **X** on the diagram, the point where the resulting electric field is zero. [7 marks]

- (b) Figure 4b shows two large parallel metal plates 10 cm apart and having a uniform electric field between them. An electron is released from the negative plate at the same time that a proton is released from the positive plate. They pass each other a distance d from the negative plate.

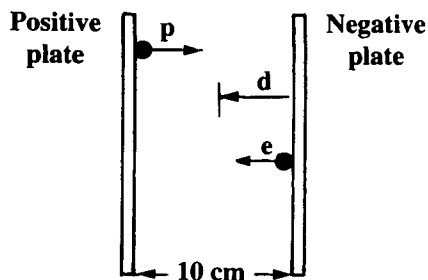


Figure 4b

- (i) Find the value of d .
- (ii) What assumption have you made to calculate the value of d ? [9 marks]

- (c) Figure 4c shows FOUR point charges arranged in a square of side 1.5 m.

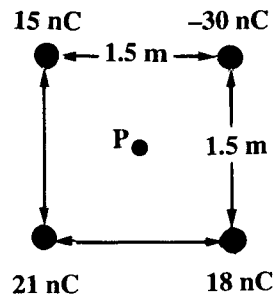


Figure 4c

Calculate the electric potential at point P , located at the centre of the square.

(Use Coulomb's constant, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$).

[4 marks]

Total 20 marks

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5. (a) Explain, with the aid of a diagram, Fleming's Left Hand Rule. [3 marks]
- (b) An electron, moving with velocity v , is about to enter a region with uniform magnetic field B . The electron moves perpendicularly to the magnetic field B as shown in Figure 5. The magnetic field is directed out of the page.

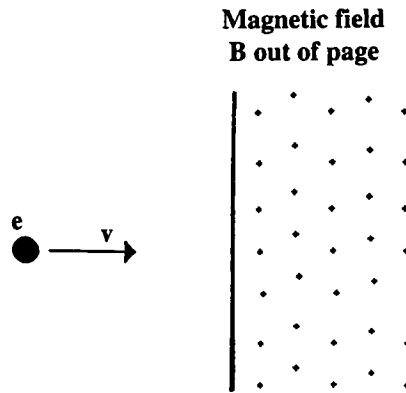


Figure 5

- (i) Copy Figure 5 into your answer booklet and draw the path of the electron as it passes through the magnetic field. On the same diagram draw the path if the electron is replaced with a proton.
- (ii) What work is done on the electron as it passes through the magnetic field? Explain your answer.
- (iii) Show that the path described by the electron in the magnetic field is a circle of radius r given by $r = \frac{mv}{Be}$ where m is the mass of the electron and e is the electron's charge.
- (iv) On the Figure 5 in your answer booklet, draw the path of the electron if the magnetic field B was reduced to half its value. [10 marks]
- (c) An electron is accelerated through an electric potential V . The electron then enters a region of constant magnetic field B oriented perpendicular to its path. In this region, the electron's path is circular with radius of curvature 10 cm. Another particle with the same charge as the electron but with charge m_x follows under the same condition. Its radius of curvature is 15 cm. Determine the mass, m_x , of the particle. [7 marks]

Total 20 marks

MODULE 2

Answer EITHER Question 6 OR Question 7.

6. (a) (i) Figure 6a shows an ideal non-inverting amplifier circuit.

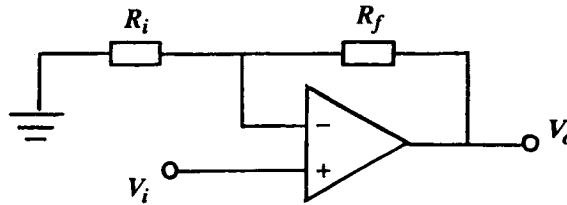


Figure 6a

Show that the closed loop voltage gain, A , in the circuit is given by $A = 1 + \frac{R_f}{R_i}$.

- (ii) How does the circuit differ from that of an inverting amplifier circuit?
- (iii) What is the input impedance of the non-inverting amplifier? [8 marks]
- (b) The saturation voltage of the non-inverting amplifier shown in Figure 6a is 15 V. If $R_f = 100 \text{ k}\Omega$ and $R_i = 50 \text{ k}\Omega$, determine the maximum input voltage, v_i , such that saturation just occurs. [2 marks]
- (c) The non-inverting amplifier circuit shown in Figure 6b is to be designed so that it can be adjusted from a gain of 20 to 50 by the rheostat R_s . The value of R_i is selected as $R_i = 1 \text{ k}\Omega$.

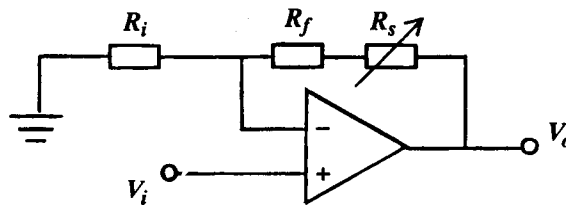


Figure 6b

- Determine the value of R_f and the range of the rheostat R_s . [6 marks]
- (d) In a certain application, it is desired to combine two signals v_1 and v_2 to form a signal v_o according to the relation $v_o = -v_1 - 20v_2$. The minimum input resistance for both signal inputs should be no less than $10 \text{ k}\Omega$. Design a circuit to meet this requirement. [4 marks]

Total 20 marks

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7. (a) (i) Construct the truth tables for the AND gate and the Exclusive –OR (EX-OR) gate.
- (ii) A student is asked to design a circuit to switch on and off a light emitting diode for inputs A and B . She designs the circuit shown in Figure 7a.

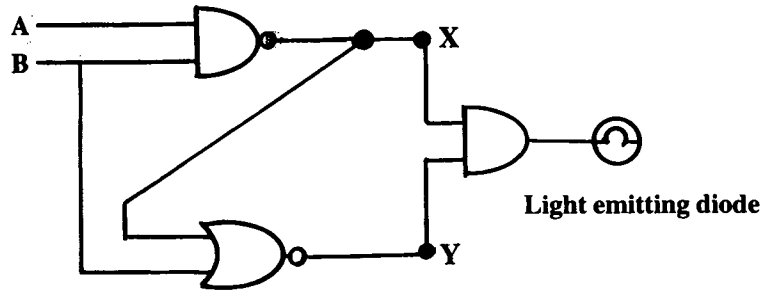


Figure 7a

Construct the truth table for the circuit and state whether or not the circuit will work.

[6 marks]

- (b) (i) Explain the operation of the Half-Adder.
- (ii) Figure 7b shows a Half-Adder circuit. Copy the waveforms A and B into your answer booklet. Under these waveforms, carefully sketch the resulting outputs S and C .

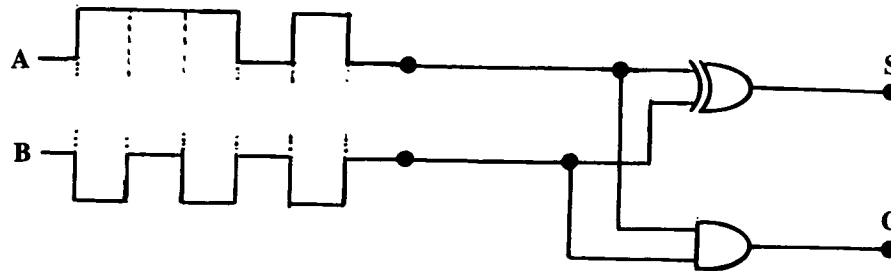


Figure 7b

- (iii) Using the circuit shown in Figure 7b, draw a circuit to implement a Full-Adder and explain its operation.
- (iv) Figure 7c shows a block diagram of a Full-Adder.

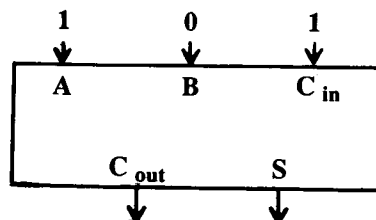


Figure 7c

If the inputs are as shown, determine the sum output, S , and carry output, C_{out} .
[14 marks]

Total 20 marks

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MODULE 3

Answer EITHER Question 8 OR Question 9.

8. (a) Explain the following terms:

- (i) Continuous x-ray spectrum
- (ii) Characteristic x-ray spectrum
- (iii) Cut-off wavelength

[8 marks]

(b) Figure 8 shows the x-ray spectrum produced when a molybdenum ($Z = 42$) target is bombarded with 35 keV electrons. The K_β and K_α wavelengths are 63.0×10^{-12} m and 70.0×10^{-12} m respectively.

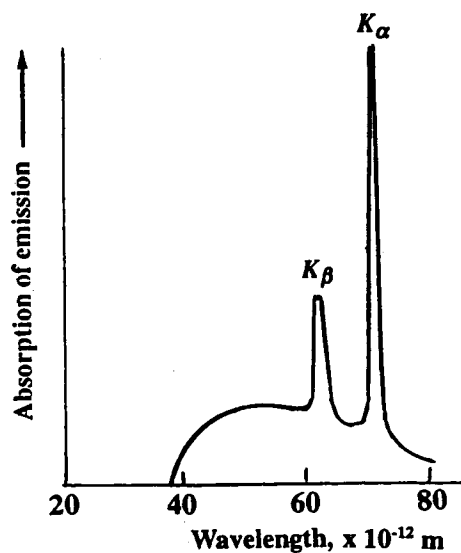


Figure 8

(i) Calculate the corresponding photon energies in eV.

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- (ii) It is desired to filter these radiations through a material that will absorb the K_β line much more strongly than it will absorb the K_α line. What substance from the table below could be used as the filtering material? Explain your answer.

The table below shows the ionization energies of the K electrons in molybdenum and in four neighbouring elements.

	Zirconium	Niobium	Molybdenum	Technitium	Ruthenium
Z	40	41	42	43	44
E_K, keV	18	18.99	20.00	21.04	22.12

- (iii) Calculate the cut-off wavelength in picometers, for molybdenum.

[12 marks]

Total 20 marks

9. (a) (i) The radioactive decay of a sample obeys the following equation:

$$N = N_0 e^{-\lambda t}.$$

Explain the meaning of the symbols in the equation and define the terms activity, A , and half-life, $t_{1/2}$.

- (ii) Show that the half-life of the sample can be written as $\frac{N \ln 2}{A}$. [8 marks]

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- (b) Figure 9 shows the results of an experiment to measure the half-life of a radioactive sample.

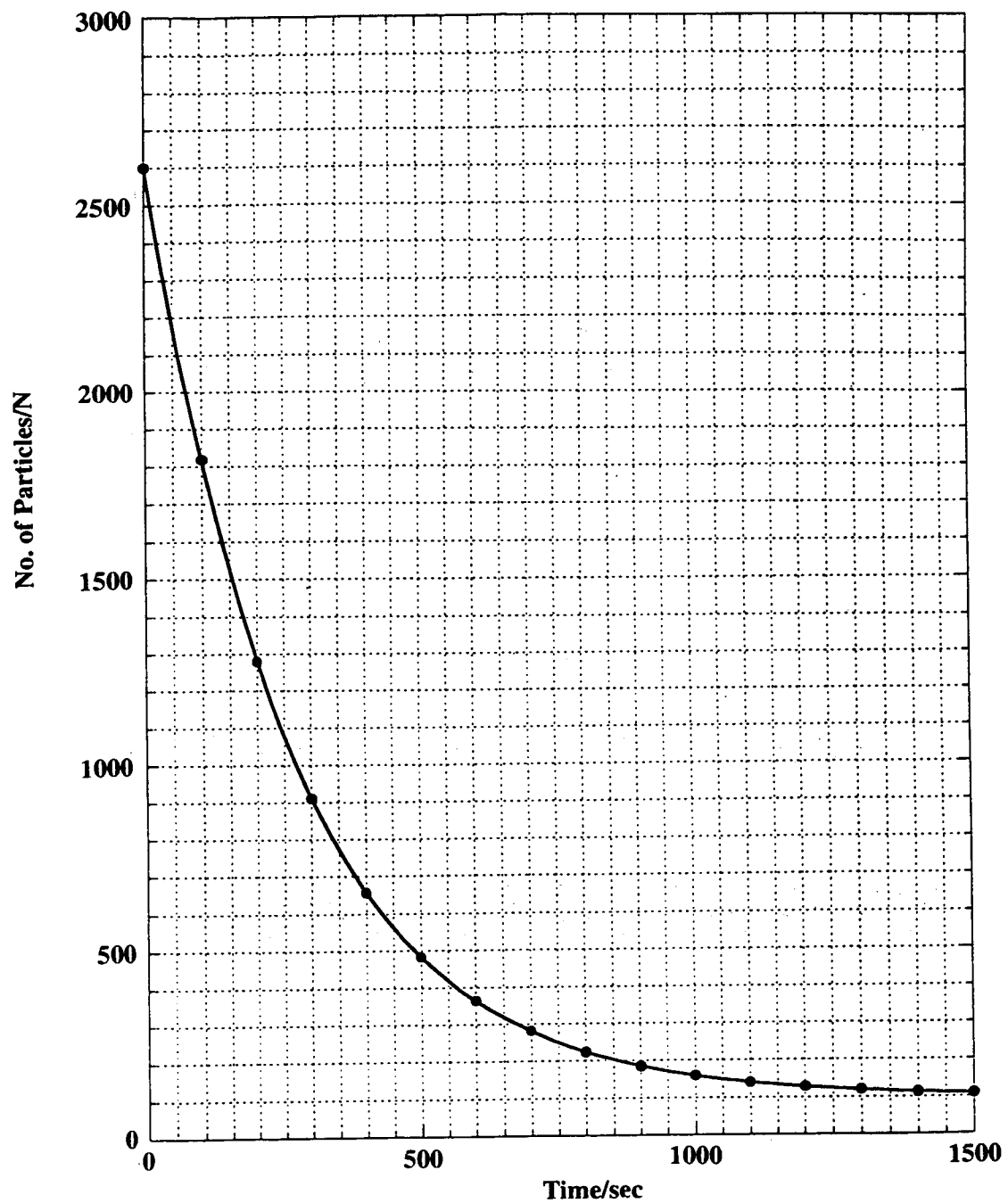


Figure 9

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Determine

- (i) the half-life of the sample
- (ii) the decay constant
- (iii) the activity at 1000 s.

[6 marks]

- (c) The thermal energy generated when radiations from radionuclides are absorbed in matter can serve as the basis for a small power source for use in satellites and remote weather stations. One such radionuclide is Plutonium, ${}^{238}_{94}\text{Pu}$ which is an alpha emitter with disintegration energy, Q , equal to 5.50 MeV. Plutonium, ${}^{238}_{94}\text{Pu}$, has a half-life of 90 years.

Calculate the rate at which thermal energy is generated in 1 kg of this material.

[Number of seconds in a year = 3.156×10^7]

[6 marks]

Total 20 marks

END OF TEST